

Blocking vs non-blocking

In a clocked always block, equals is blocking assignment and less-than-or-equals is non-blocking

Same edge, different story

- Picture a two-stage pipeline: q1 gets d, then q2 gets q1
- With blocking
- With non-blocking, q2 still sees the old q1, so you get a real one-cycle delay per stage
- The swap gotcha is the same idea
- Rule of thumb

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Blocking vs non-blocking

INTERACTIVE TOOL

Blocking vs non-blocking

Same clock edge, two always blocks: blocking = updates immediately; non-blocking <= schedules updates for the end of the time step. Watch the registers diverge side by side.

Starter example: Register swap with a=1, b=0. One $\nearrow$ posedge — blocking loses the 1; non-blocking swaps.

Load starter example

Challenges0 / 20 cleared

Blocking loses a bit

Register swap

Non-blocking swaps

Register swap

Swap again

Register swap

Blocking loses a bit: On Register swap, take one $\nearrow$ posedge. Blocking side should show a=0 and b=0 (the 1 is gone).

Start

Show hint

Next

Stop

Idle

Scenario

Register swap

Try to swap a and b in one always block — classic gotcha.

Two-stage pipeline

$q1 \leftarrow d, q2 \leftarrow q1$ — one cycle delay only with <=.

Blocking vs non-blocking lab starter

Real Verilog practice

```
wsl - module08-blocking-vs-nonblocking/examples/navigation

# Track A - real Unix
# real Verilog session (Track A)

$ cat examples/blocking_nba/blocking_nba.v
// blocking_nba.v - sequential style uses non-blocking
module ff_nba(
    input    clk,
    input    D,
    output reg Q
);
    always @(posedge clk)
        Q ≤ D;
endmodule

module pipe_nba(
    input    clk,
    input    D,
    output reg q1,
    output reg q2
);
    always @(posedge clk) begin
        q1 ≤ D;
        q2 ≤ q1; // old q1 - true pipeline delay
    end
endmodule

$ iverilog -t null examples/blocking_nba/blocking_nba.v
(syntax check passed)
```

Real Verilog — flip-flop and pipeline with non-blocking

Pitfalls to watch

- In coursework you usually do not
- Do not mix blocking and non-blocking on the same register in one edge without a clear
- Do not use non-blocking in combinational at-star blocks, it is for sequential logic
- If your pipeline shows zero delay between stages, check for blocking assignments first

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On real Verilog
- When you are ready